

Jichen YAO

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EDUCATION

Northeastern University

Bachelor's Degree

Sep 2023 – Jul 2027 (Expected)

Liaoning, China

- Major: Computer Science and Technology | GPA: 3.89/5.0
- Core Courses: Fundamentals of Programming (C Language), Object-Oriented Programming (C++), Intelligent Electronics and Information Technology, Algorithm Design and Analysis, Numerical Analysis, Digital Logic and Digital Systems, Basic Training of Information Technology, Data Structures, Probability Theory and Mathematical Statistics, Linear Algebra, Advanced Mathematics, Discrete Mathematics

RESEARCH PROJECTS

Fractal Spring Pyramid Memory Architecture for Long Context in LLMs

Mar 2026 – Present

Advisor: Assoc. Prof. Shi Feng

- Investigated how hierarchical memory mechanisms can improve long-context modeling by retaining salient information beyond local attention windows.
- Designed a Fractal Spring-style memory pyramid with Micro, Meso, and Macro levels, using adaptive gates to control memory selection, update, and injection.
- Built an initial experimental pipeline for memory distillation and long-context retrieval evaluation; currently refining the architecture through empirical validation.

Cognitive-Inspired Strategy Optimization for Emotional Support Dialogue (ESC)

Dec 2025 – May 2026

Advisor: Assoc. Prof. Shi Feng

- Developed CogCSO, a cognitive-structure-aware preference optimization framework for ESC, encoding five therapeutic pacing principles into contrastive data construction to mitigate premature problem-solving.
- Extended the CSO-style MCTS preference construction pipeline by introducing a Cognitive Alignment reward signal, encouraging validation-before-intervention pacing during response selection
- Constructed dialogue preference pairs from reward-guided MCTS rollouts and fine-tuned LLM backbones with LoRA and DPO using the LLaMA-Factory framework.
- Designed a robust evaluation framework (point-wise, pairwise blind, statistical tests) to analyze automated metric instability. Identified and analyzed the Verbosity Tax, validating it through cross-model contrast (LLaMA vs. Qwen).

Aspect-aware Multi-head Cross-Attention for Aspect-based Sentiment Classification

Jul 2025 – Nov 2025

Advisor: Assoc. Prof. Shi Feng

- Proposed AMCA, an end-to-end model leveraging learnable aspect embeddings and multi-head cross-attention to extract semantics dynamically without external NLP tools.
- Redefined ABSA task objectives by shifting the focus from marginal classification accuracy to aspect detection completeness.
- Achieved state-of-the-art performance in aspect detection tasks across multiple datasets, demonstrating significant advantages in identifying implicit aspects through independent model design and systematic empirical validation.
- **Publication:** Jichen Yao, Shi Feng. "Aspect-aware Multi-head Cross-Attention for Aspect-based Sentiment Classification." *Accepted by ICNC-FSKD 2026.*

EXTRACURRICULAR ACTIVITY

Dalian Municipal Culture and Tourism Bureau Warm Winter Volunteer Service

Jan 2025

Role: Volunteer | Organizer: Dalian Municipal Culture and Tourism Bureau

- Provided on-site assistance and information services to returning travelers at major transportation hubs including airports and railway stations.
- Delivered customer-oriented solutions to address inquiries regarding transportation, luggage, and local tourism information.
- Collaborated with a team of volunteers to ensure smooth operation of winter travel assistance initiative.

TECHNICAL SKILLS

- Languages: Mandarin (Native), English (TOEFL iBT Home Edition: 105)
- Programming: Proficient in Python, C, C++; Familiar with Shell scripting
- AI/ML Stack: PyTorch, LLaMA-Factory, HuggingFace, MCTS, DPO, Parameter-Efficient Fine-Tuning (LoRA)
- Tools: Git, Docker, LaTeX, Visual Studio Code, Linux Environment